



Indian Institute of Technology, Guwahati

Guwahati, INDIA 781 039

Department of Chemistry

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CH101

Classes 3; Physical Chemistry

Schrödinger Wave Equation

Time –dependent Schrödinger Wave Equation

$$\hat{H}\Psi(r,t) = i\hbar \frac{\partial \Psi}{\partial t}$$

Time – independent Schrödinger Wave Equation

$$\hat{H}\Psi = E \Psi$$

Ψ is a function of spatial coordinates (x, y, z) or (r, θ , ϕ) for a particular energy state (E).

$\hat{H}$ is known as the Hamiltonian operator or simply Hamiltonian.

$\hat{H}$ = (Kinetic Energy + potential Energy) operators = $\hat{T} + \hat{V}$

Typically, $\hat{T} = \frac{p^2}{2m} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$ in three dimensions

In one dimension, $\hat{T} = \frac{p_x^2}{2m} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$

$\hat{V}$ depends on the system and interaction involved.

For Coulombic potential: $\hat{V} = -\frac{Ze^2}{4\pi\epsilon_0 r}$ (between an electron and nucleus with charge Z)

For Harmonic Oscillator potential: $\hat{V} = \frac{1}{2} kX^2$

Solving Problems: Particle in a BOX. One-Dimensional Box. The particle is an electron and it does not have enough energy to climb the barrier, therefore remains confined to the box. What are the energy states that the particle can occupy?

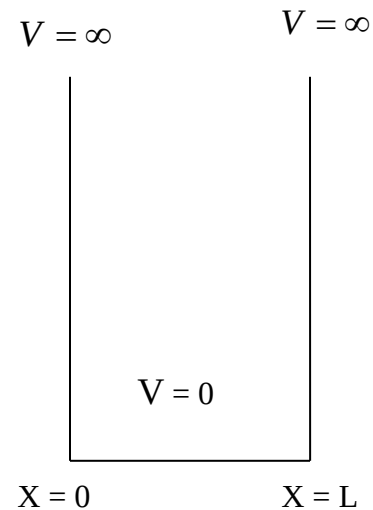
Hamiltonian, $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$

$$\hat{H}\Psi = E \Psi$$

Or, $-\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} = E \Psi$

Or, $\frac{d^2\Psi}{dx^2} = -\frac{2mE}{\hbar^2} \Psi = -k^2 \Psi;$

where $k = \frac{(2mE)^{1/2}}{\hbar}$



Solutions: $\Psi = Ae^{ikx}; \Psi = Be^{-ikx}$ and $\Psi = Ae^{ikx} + Be^{-ikx}$

However, $\Psi = Ae^{ikx} + Be^{-ikx}$ is the most general solution.

$\Psi = Ae^{ikx} + Be^{-ikx}$, however, can be rewritten as $\Psi = C \cos kx + D \sin kx$

Boundary conditions:

At $x = 0$; $\Psi = 0$ and at $x = L$; $\Psi = 0$.

When $x = 0$; $\Psi = 0$; then $\Psi = C \cos kx + D \sin kx = C = 0$

Hence, $\Psi = D \sin kx$

However, at $x = L$; $\Psi = 0$; then $\Psi = D \sin kL = 0$

This is possible only when $kL = n\pi$; where $n = 1, 2, 3, 4, \dots$

Hence, $\Psi = D \sin \frac{n\pi x}{L}$; the solution of the equation under the above boundary conditions.

One can also write $\Psi(x) = D \sin \frac{n\pi x}{L}$

However, D is still unknown. How to find D ?

Since the particle must be inside the box, the total probability of finding the particle inside the box must be unity.

Thus

$$\int_0^L \Psi^*(x) \Psi(x) dx = 1$$

$$\int_0^L D \sin \frac{n\pi x}{L} D \sin \frac{n\pi x}{L} dx = 1$$

$$D = \sqrt{\frac{2}{L}}$$

Hence, $\Psi(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$; complete solution.

One can see the solution is expressed in terms of the dimension of the box (L), position (x) and quantum numbers (n=1, 2, 3, ...).

One can derive the expression for energy of the particle in the box from

$$-\frac{\hbar^2}{2m} \frac{d^2 \Psi}{dx^2} = E \Psi$$

With the wavefunction $\Psi(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$;

The expression for energy is

$$E_n = \frac{n^2 \hbar^2}{8mL^2}; \text{ where } n \text{ is the quantum numbers, } n = 1, 2, 3, 4, \dots$$

Home work:

Find the average position of the particle in the one-dimensional box with the length L.

Hint: Evaluate the following integral

$$\langle x \rangle = \int_0^L \Psi(x)^* x \Psi(x) dx$$

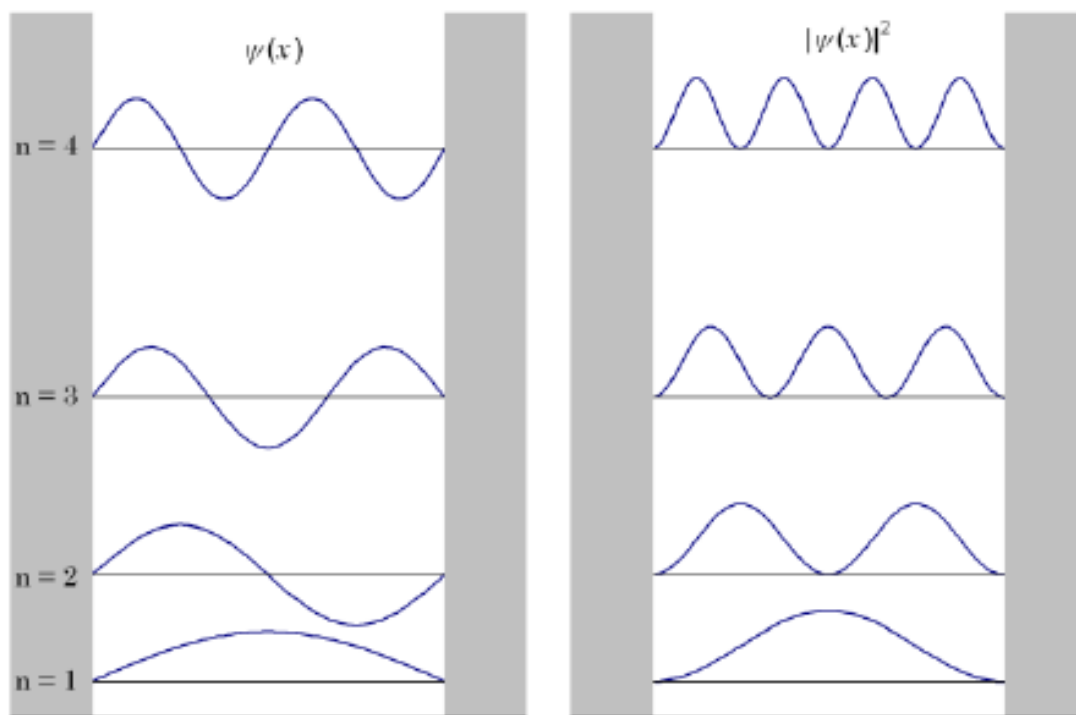


Figure 1. Wavefunctions and probability densities of the first four states of particle in a one-dimensional box.